

Chapter 6

Object Reconstruction and Identification

Hadronically decaying τ -leptons, electrons, muons and jets are used in this analysis. This chapter describes their reconstruction, identification and calibration, followed by their definitions in the analysis, overlap removal and the reconstruction of E_T^{miss} .

These objects are reconstructed and identified using detector measurements. Tracks are used to determine charged-particle momenta and production vertices, while energy deposits are measured in the calorimeters. The reconstructed energies and momenta are calibrated to correct biases in the detector response. Simulated event weights are further corrected using efficiency scale factors.

6.1 Tracks

6.1.1 Reconstruction

A track is reconstructed by fitting the trajectory of a charged particle to hits in the ID. Its momentum and charge are determined from the curvature in the magnetic field, while its position near the beam line is used to assign it to a production vertex. Track reconstruction is performed in the following stages [1]:

1. **Space-point formation.** Signals in neighbouring channels of a silicon sensor are grouped into clusters. The coordinates provided by Pixel clusters and SCT clusters are converted into three-dimensional space points.
2. **Seeding and track finding.** Combinations of space points are used to form track seeds. These seeds are extended through the silicon layers with a combinatorial Kalman filter, which is used to refine the track parameters.

3. **Ambiguity resolution.** Track candidates are ranked using their fit quality, momentum and hit variables. Meanwhile, candidates that share the same clusters are marked as "shared hits". Candidates fail the track-quality requirement are then rejected to suppress duplicate and fake tracks.
4. **TRT extension.** Tracks are extended with additional TRT hits. The extended track is retained if its quality score is higher than that of the original track reconstructed using only pixel and SCT hits.

6.1.2 Track parameters and quality

The fitted trajectory is described by five parameters, as illustrated in Figure 6.1:

- d_0 : distance in the $x - y$ plane between the beam line and the perigee point, defined as the point on the track closest to the beam line.
- z_0 : the z coordinate of the perigee point relative to the reference plane at $z = 0$.
- ϕ and θ : the azimuthal and polar angles of the momentum, respectively.
- q/p : the electric charge divided by the momentum magnitude.

Two quantities are used to test whether a track is consistent with production at the primary vertex [2]:

- **Transverse impact-parameter significance**, d_0/σ_{d_0} : the transverse distance from the beam line in units of its uncertainty σ_{d_0} .
- **Projected longitudinal distance**, $\Delta z_0 \sin \theta = (z_0 - z_{PV}) \sin \theta$: the longitudinal distance between the track perigee and the primary vertex. Here, z_{PV} is the z coordinate of the primary vertex.

Small absolute values of both quantities suggest that the track is from the primary vertex. A large longitudinal distance can also indicate a track from another pp collision. Prompt electron and muon candidates are therefore selected by requiring small values of both quantities, and displaced candidates with large values can originate from a separate decay vertex, for example from a b - or c -hadron decay.

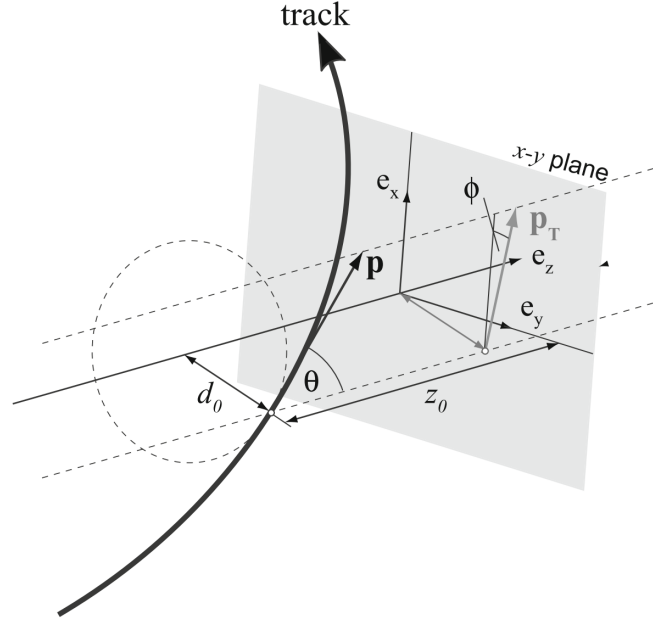


Fig. 6.1 Geometry of the Track parameters. The grey plane represents the reference $z = 0$ plane [3].

6.2 Primary vertex

A primary vertex (PV) is the reconstructed position of a pp collision. It is used as the reference for prompt-lepton selection, isolation, pile-up jet rejection and the track contribution to E_T^{miss} . Its position is determined from tracks originating from a common point. Several PVs can be reconstructed in an event because of pile-up. The vertex with the largest sum of squared transverse momenta of its tracks is selected as the PV of the hard scattering [4].

6.3 Electrons

6.3.1 Reconstruction

The electron object is reconstructed from an ID track matched to an electromagnetic shower in the calorimeter. Bremsstrahlung photons emitted in the upstream material can spread the shower energy over several clusters and alter the curvature of the electron track. Both effects are treated in the reconstruction [5]:

1. **Topo-cluster reconstruction.** Neighbouring cells are grouped into topological clusters according to their signal significance above the expected noise. Clusters with a substantial electromagnetic energy fraction are used as electron seeds.

2. **Track refitting and matching.** Electron-track candidates are refitted with a Gaussian-sum filter, in which a mixture of Gaussian components is used to describe radiative energy losses. The refitted tracks are extrapolated to the second layer of the EM calorimeter and matched to clusters using their differences in η and ϕ . If several tracks match the same cluster, the ΔR between each extrapolated track position and the cluster centre is used together with the pixel and SCT hits to rank the tracks.
3. **Supercluster formation.** Nearby clusters are added to the seed to recover energy from bremsstrahlung. Additional clusters sharing the matched electron track can be collected over a wider region in ϕ .

The cluster association is illustrated in Figure 6.2. The wider collection in ϕ helps recover energy separated from the electron by its bending in the magnetic field.

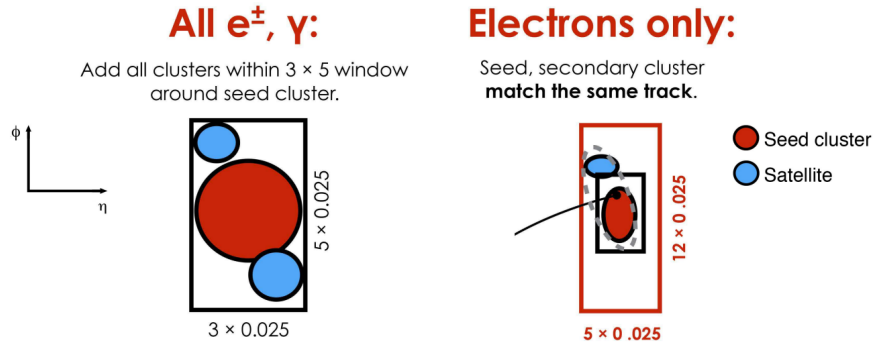


Fig. 6.2 Cluster association for electron superclusters. Seed and additional clusters are shown in red and blue, respectively. Geometrically nearby clusters are collected [5].

6.3.2 Identification

Electron candidates are distinguished from hadronic showers and converted photons using a likelihood discriminant [6]. Its inputs include the shower shapes in the EM calorimeter, hit variables, and the agreement between the track and calorimeter measurements. The Loose, Medium and Tight working points are defined and provide increasing background rejection with the different likelihood discriminant threshold. The Tight likelihood working point, denoted TightLH, is used for all electron candidates in this analysis. Its typical prompt-electron identification efficiency is about 80% [7].

6.3.3 Isolation

Electrons from hadron decays can pass the electron identification requirements. These electrons are usually accompanied by other particles from the same jet, while prompt electrons typically have smaller nearby track p_T and calorimeter E_T sums. Isolation requirements are therefore used to suppress this background.

Two isolation variables are calculated using tracks and calorimeter clusters within cones around the electron direction [5]:

- **Track isolation** is performed using the p_T sum of tracks in the cone. At high electron p_T , other decay products can lie close to the electron due to the boost effect, so the cone radius is reduced.
- **Calorimeter isolation** follows the same principle as track isolation, but uses cluster E_T within a fixed cone of $\Delta R = 0.2$. The corrected sum estimates the E_T from other particles near the electron.

In the analysis, baseline electrons are required to pass Loose_VarRad isolation while signal electrons must additionally satisfy HighPtCaloOnly. The Loose_VarRad working point requires the track and calorimeter isolation sums to be below 15% and 20% of the electron p_T , respectively. The HighPtCaloOnly working point requires the calorimeter isolation sum to be below $\max(0.015 p_T, 3.5 \text{ GeV})$. At high E_T , the prompt-electron efficiencies are about 99% for Loose_VarRad and 90–95% for HighPtCaloOnly [7].

6.3.4 Calibration

The electron energy is calibrated for losses in upstream material, energy outside the cluster and variations in detector response [5]. $Z \rightarrow ee$ events are used for the calibration. Reconstruction, identification and isolation efficiencies are measured with the tag-and-probe method: one electron (the tag) is required to pass strict identification cuts, while the other is used as the probe. After background subtraction, the efficiency is the fraction of probes that pass the selection. For a selection efficiency ε , the scale factor is defined as

$$\text{SF} = \frac{\varepsilon_{\text{data}}}{\varepsilon_{\text{MC}}}. \quad (6.1)$$

The scale factors are then applied to simulated event weights for the electron working points used in this analysis.

6.4 Muons

6.4.1 Reconstruction

Muons are reconstructed using tracks in the ID and MS, together with their energy deposits in the calorimeters [2]. The main reconstruction categories are:

- **Combined muons** are reconstructed by matching and fitting ID and MS tracks, including the energy loss between the two tracking systems.
- **Segment-tagged muons** are reconstructed by matching an ID track to one or more local track segments in the MS when a complete MS track is unavailable.
- **Calorimeter-tagged muons** are reconstructed by matching an ID track to calorimeter deposits compatible with a minimum-ionising particle. Acceptance is recovered in regions with limited MS coverage.
- **MS-extrapolated muons** are reconstructed by extrapolating an MS track towards the interaction region without a matched ID track.

Combined muons are used in this analysis.

6.4.2 Identification

Muon identification tests whether the detector hits and the ID and MS momentum measurements are consistent with a single muon trajectory [2]. Decays of charged mesons in flight can change the trajectory and momentum, leading to a disagreement between the momentum measurements. For muon candidates, this agreement is quantified by the q/p significance,

$$S_{q/p} = \frac{|(q/p)_{\text{ID}} - (q/p)_{\text{MS}}|}{\sqrt{\sigma_{\text{ID}}^2 + \sigma_{\text{MS}}^2}}, \quad (6.2)$$

where $(q/p)_{\text{ID}}$ and $(q/p)_{\text{MS}}$ are the charge-to-momentum ratios determined from the ID and MS tracks, respectively. σ_{ID} and σ_{MS} denote the uncertainties. Large values indicate poor agreement and help reject incorrectly matched candidates.

In the analysis, baseline muons are required to pass Loose identification while signal muons must additionally satisfy the Medium working point. The Medium working point requires combined candidates within the ID acceptance to satisfy $S_{q/p} < 7$ and, outside $|\eta| < 0.1$, have at least two MS precision stations with at least three MDT or CSC hits each. The Loose working

point applies the same requirements and additionally accepts segment-tagged and calorimeter-tagged candidates in $|\eta| < 0.1$, where MS coverage is limited. The typical prompt-muon efficiencies are about 99% for Loose and 97% for Medium.

6.4.3 Isolation

Muons from heavy-flavour decays are often accompanied by other particles inside a jet. They are suppressed using track and calorimeter isolation, defined by the surrounding activity after excluding the muon's own contribution [2]. Baseline muons are required to pass Loose_VarRad isolation, and signal muons must additionally satisfy Tight_VarRad. Both working points use a track-cone radius that decreases as the muon p_T increases and a fixed calorimeter cone of $\Delta R = 0.2$. The track and calorimeter isolation sums must be below 15% and 30% of the muon p_T for Loose_VarRad, respectively. These limits are reduced to 4% and 15% for Tight_VarRad, allowing less activity around the muon and rejecting more muons produced inside jets. The typical prompt-muon isolation efficiencies are about 95% for Loose_VarRad and 86% for Tight_VarRad [8].

6.4.4 Calibration

The muon momentum scale and resolution are calibrated using resonance decays, e.g., $Z \rightarrow \mu\mu$ and $J/\psi \rightarrow \mu\mu$ [9]. The reconstructed mass distributions constrain biases in the muon momentum measurement. The efficiency scale factors are measured with tag-and-probe samples as for electrons and applied to simulated event weights for the selected working points [2].

6.5 Jets

6.5.1 Reconstruction

Jets are reconstructed from the collimated hadrons produced through parton showering and hadronisation. In this analysis, the constituents used for jet clustering are first reconstructed with the particle-flow (PFlow) algorithm, which combines ID track momenta with calorimeter energy measurements [10]. The main steps are:

1. Calorimeter topological clusters are initially calibrated for electromagnetic showers and matched to ID tracks.
2. The expected calorimeter energy deposited by the matched charged particles is subtracted from the clusters, avoiding double counting when the track momenta are used.

3. Vertex association is used to suppress charged-particle contributions from pile-up. The selected tracks and remaining calorimeter contributions form the PFlow constituents.

These PFlow constituents are then grouped into jets using the anti- k_t algorithm with $R = 0.4$ [11]. Pairs are selected for combination according to their angular separation and p_T . Their four-momenta are added, and the merged object is used in the next clustering step. Low- p_T constituents are preferentially clustered around nearby high- p_T constituents, so isolated jets have approximately circular boundaries with radius R .

6.5.2 Identification and pile-up rejection

Jets from the hard scattering are selected after jet cleaning and pile-up rejection.

- **Jet cleaning** is performed with the requirements on calorimeter pulse shapes and the distribution of energy among calorimeter layers to reject noise and non-collision backgrounds. For example, a jet with $|\eta| < 2$ is rejected if more than 99% of its energy is deposited in a single calorimeter layer. The efficiency for collision jets exceeds 99.5% [12].
- **Pile-up rejection** uses the vertex origins and p_T of the jet tracks. The jet-vertex tagger (JVT) is constructed from two variables [13]. The first is based on the ratio of the p_T sum of tracks from the PV to that of tracks from all vertices. The second is the same PV-track p_T sum divided by the reconstructed jet p_T . Large values of both variables favour jets from the hard scattering. The neural-network JVT (NNJVT) is used in this analysis with the FixedEffPt working point [14], whose target efficiency for hard-scatter jets is 88–98% [15].

6.5.3 Calibration

Jet energies are calibrated to correct the jet energy scale (JES) and improve the jet energy resolution (JER). The JES characterises the average energy response, while the JER quantifies the spread of reconstructed energies at a fixed particle-level energy. The following corrections are applied [16]:

1. **Pile-up subtraction** removes the estimated energy deposited by pile-up particles. The corresponding correction to the jet p_T is estimated as ρA , where ρ is the event p_T density and A is the jet area.
2. **Energy and direction calibration** corrects the differences in energy and η between reconstructed and matched particle-level jets. Particle-level jets are built from stable

simulated particles, excluding muons and neutrinos, and therefore provide a reference without detector effects.

3. **Global sequential calibration** corrects the remaining response differences between jets with different particle compositions or shower shapes. Track multiplicity and energy fractions in calorimeter layers are used to reduce these differences, improving the resolution while preserving the average energy scale.

4. **Residual in situ calibration** uses well-measured Z bosons and photons as references to check the jet energy response. For higher- p_T jets, the vector sum of several calibrated lower- p_T jets are used as the reference. The average ratio of jet p_T to reference p_T is compared between data and simulation to determine the final correction, and then applied only to data.

6.6 Jet flavour tagging

Jet flavour tagging is used to distinguish b -quark originating jets (b -jets) from other jet flavours. b -jets are identified through displaced b -hadron decays, which have a lifetime of about 1.5 ps. Therefore a boosted b -hadron can travel several millimetres before decaying. The decay tracks therefore typically have large impact parameters and originate from a secondary vertex displaced from the PV. Charm hadrons produced in the b -hadron decay can further decay at an additional displaced vertex. These features are used for jet flavour tagging.

The tagger uses the GN2v01 algorithm, which is based on a transformer neural network trained on simulated events [17]. Its inputs include jet and track kinematics, track impact parameters, and hit variables. These variables are first encoded as a numerical vector for each track. Within the transformer, each vector is updated through self-attention, using information from the other tracks. Correlations between tracks can thereby be used to recognise a common displaced origin without requiring explicitly reconstructed secondary vertices as separate inputs. The network then calculates a weight for each track using its updated vector and adds the weighted vectors together, which is called attention pooling.

The resulting vector is used for the main task of jet-flavour classification. It is also combined with the individual track vectors for two auxiliary tasks: identifying track origins and predicting whether pairs of tracks originate from the same vertex. The three tasks share the transformer and are trained jointly, helping it learn the decay structure. The network and its three tasks are illustrated in Figure 6.3.

The jet-flavour classifier produces four outputs for the b -jet, c -jet, hadronic- τ and light-jet hypotheses, where light jets originate from u , d or s quarks or gluons. These outputs are

combined into a single discriminant, D_b , which compares the b -jet output with a weighted combination of the other three outputs. Larger values favour the b -jet hypothesis. The GN2v01 85% working point is used in this analysis, corresponding to a nominal b -jet tagging efficiency of 85%.

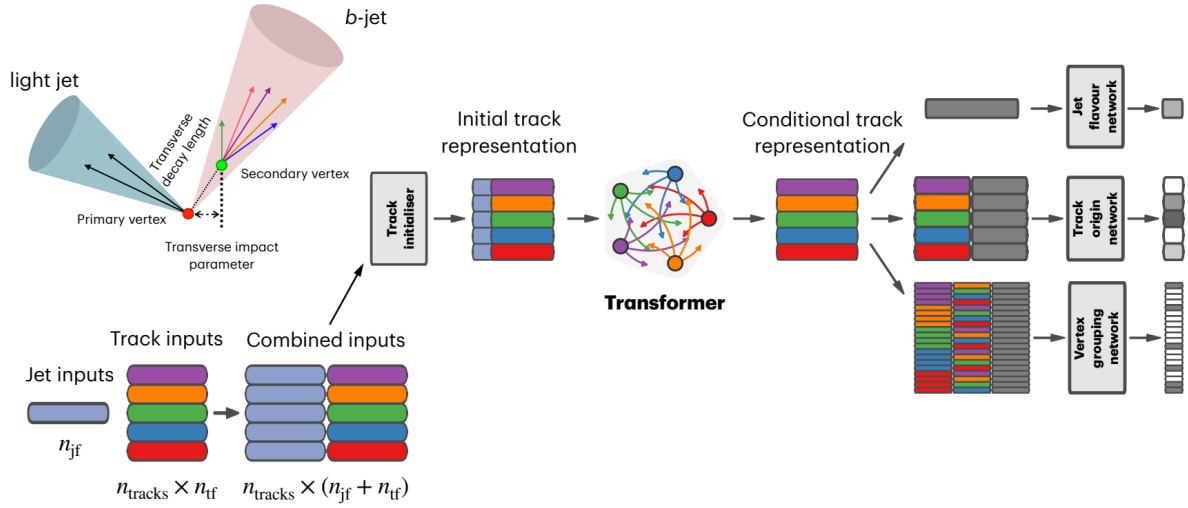


Fig. 6.3 GN2 flavour-tagging architecture. The jet variables are appended to the input vector of each track before processing by the track initialiser and transformer. The resulting track vectors and their weighted combination are passed to the jet-flavour, track-origin and vertex-grouping networks. The symbols n_{jf} , n_{tf} and n_{tracks} denote the numbers of jet variables, track variables and tracks, respectively [17].

6.6.1 Calibration

The b -jet tagging efficiency is measured in dileptonic $t\bar{t}$ events with an oppositely charged electron–muon pair and two jets [17]. The c -jet mistagging efficiency, the probability for a c -jet to pass the b -tagging requirement, is measured in semileptonic $t\bar{t}$ events using charm jets from hadronic W decays. The corresponding light-jet mistagging efficiency is measured in Z +jets events with $Z \rightarrow ee$ or $Z \rightarrow \mu\mu$ decays.

These efficiencies are measured in jet p_T bins and compared with simulation to obtain flavour-dependent scale factors, as defined in Eq. 6.1. Corrections are then applied to the simulated event weights for both tagged and untagged jets.

6.7 Hadronic tau leptons

6.7.1 Reconstruction

τ leptons can decay both leptonically and hadronically. Approximately 65% of τ decays are hadronic, with the major modes containing charged and possible neutral pions [18]. In this analysis we only consider the hadronic decay tau. It is denoted by τ_{had} . Here the τ -neutrino is excluded in the reconstruction.

Hadronic tau candidates are first seeded by anti- k_t jets with $R = 0.4$. The production vertex is then chosen as the vertex with the largest fraction of the p_T sum of tracks within $\Delta R < 0.2$ of the seed axis. This vertex provides the reference for track association and impact-parameter calculations. The tracks are then classified by a recurrent neural network (RNN) to select tau-decay tracks. The number of selected tau-decay tracks defines the prong multiplicity: candidates with one or three such tracks are called one-prong or three-prong candidates, respectively.

6.7.2 Identification

Hadronic tau decays are identified using an RNN discriminant to distinguish them from quark- and gluon-initiated jets [19, 18]. Track momenta and calorimeter shower properties are used as inputs, since tau decays typically have narrower showers and lower charged multiplicity than these jets. Track displacements and, for three-prong candidates, the significance of the reconstructed decay-vertex displacement are also included as inputs sensitive to the finite tau lifetime. Nearby activity is also included in the RNN inputs, so isolation information is used as part of the identification.

Baseline tau candidates are required to pass Loose RNN identification, and signal candidates must additionally satisfy Tight. The higher RNN-score threshold for Tight rejects more jets at the cost of tau efficiency. For reconstructed one-prong (three-prong) candidates, the target identification efficiencies are 85% (75%) for Loose and 60% (45%) for Tight.

Electrons can be misidentified as hadronic tau candidates because their narrow showers and tracks are similar to the signature of a one-prong tau decay [18]. To suppress this background, both baseline and signal tau candidates are required to pass Loose electron rejection. This working point requires a minimum score from a separate RNN using shower and track information, including the relation between calorimeter energy and track momentum. The tau efficiencies are approximately 95% and 98% for one-prong and three-prong candidates, respectively. The corresponding electron rejection efficiencies are approximately 99.7% and 98.8%.

6.7.3 Calibration

The tau energy scale (TES) calibration is derived from MC samples [18]. In these samples, the generated visible tau p_T , excluding the decay neutrino, is used as the reference for correcting the reconstructed p_T . The reconstructed p_T is measured both from calorimeter deposits alone and from a combination of charged-hadron track momenta and neutral-particle calorimeter deposits. These two measurements are combined in a weighted average. The averaged p_T is then corrected using a boosted regression tree trained on these MC samples.

In addition to the momentum calibration, differences in tau reconstruction, identification and electron-rejection efficiencies between data and simulation must be corrected. For this purpose, data-to-simulation scale factors are applied to simulated event weights. In this analysis, for signal candidates passing Tight tau identification, the electron-veto scale factors for Medium tau identification are used because no corresponding factors were available for Tight.

6.8 Overlap removal

Different reconstructed objects can share the same detector measurements. ΔR is calculated using rapidity for overlap removal. An ordered overlap-removal procedure is therefore applied to resolve these ambiguities before event selection. Overlap removal is applied to the baseline objects in the order listed below.

1. **Electron–electron:** the lower- p_T electron is removed if two electrons share an ID track, so that a single track is not counted as two electrons.
2. **Tau–electron:** a tau candidate within $\Delta R < 0.2$ of an electron is removed, since an electron shower and its track can also be reconstructed as a tau candidate.
3. **Tau–muon:** a tau candidate within $\Delta R < 0.2$ of a muon is removed to reject muons misidentified as hadronic tau decays.
4. **Calorimeter-tagged muon–electron:** a calorimeter-tagged muon sharing the electron ID track is removed. The electron interpretation is preferred because the calorimeter tag has no independent MS confirmation.
5. **Electron–muon:** an electron sharing an ID track with a remaining muon is removed.
6. **Jet–electron:** a jet within $\Delta R < 0.2$ of an electron is removed because the electron’s EM shower can also be reconstructed as a jet.

7. **Electron–jet:** an electron within $\Delta R < 0.4$ of a remaining jet is removed to suppress electrons from hadron decays inside jets.
8. **Jet–muon:** a jet with fewer than three tracks is removed if the muon is ghost-associated with it.¹
9. **Muon–jet:** a muon within $\Delta R < 0.4$ of a remaining jet is removed to suppress muons from hadron decays inside jets.
10. **Jet–tau:** a jet within $\Delta R < 0.2$ of a remaining tau candidate is removed, so that the visible tau-decay products are not also counted as a separate jet.

6.9 Missing transverse momentum

The incoming partons at the LHC carry little transverse momentum, so the vector sum of the transverse momenta of all the particles is expected to be approximately zero. An imbalance in the visible transverse momentum can therefore indicate undetected particles, such as neutrinos or weakly interacting BSM particles. In this analysis, the neutrino produced in the LQ process and the tau-decay neutrino can together give rise to sizeable missing transverse momentum ($\vec{p}_T^{\text{miss}}$). $\vec{p}_T^{\text{miss}}$ is reconstructed as the negative vector sum of the calibrated visible contributions, with its magnitude denoted by E_T^{miss} [20]:

$$\vec{p}_T^{\text{miss}} = - \left(\sum_e \vec{p}_T^e + \sum_\mu \vec{p}_T^\mu + \sum_{\tau_{\text{had}}} \vec{p}_T^\tau + \sum_j \vec{p}_T^j + \vec{p}_T^{\text{soft}} \right), \quad (6.3)$$

$$E_T^{\text{miss}} = \left| \vec{p}_T^{\text{miss}} \right|. \quad (6.4)$$

Two types of contribution are included in the reconstruction:

- **hard term:** The electron, muon, hadronic-tau and jet contributions selected for $\vec{p}_T^{\text{miss}}$ reconstruction. The jet contribution is selected using the Tight $\vec{p}_T^{\text{miss}}$ working point to suppress pile-up jets. This requires jet $p_T > 20$ GeV for $|\eta| < 2.5$ and $p_T > 30$ GeV for $2.5 \leq |\eta| < 4.5$.
- **track soft term:** The vector sum of transverse momenta of selected PV tracks not used in the hard term.

¹Ghost association means that a copy of the muon’s ID track is made, keeping its direction but reducing its momentum to a negligible value. This “ghost” is clustered together with the normal jet inputs. If the clustering algorithm places the ghost in a particular jet, the muon is associated with that jet.

6.10 Object definitions in the analysis

Baseline and signal objects are defined to provide two levels of selection in the analysis. After overlap removal as described in Section 6.8, baseline objects are used for object counting and lepton vetoes. Signal objects are selected from these baseline collections with tighter identification or isolation requirements, as summarised in Table 6.1.

Table 6.1 Baseline object definitions and additional signal requirements used in the analysis. All baseline requirements are retained for signal objects; a dash denotes no additional requirement.

Object	Requirement	Baseline	Additional signal requirement
Electron	p_T	$> 10 \text{ GeV}$	—
	$ \eta $	< 1.37 or $1.52 < \eta < 2.47$	—
	$ d_0 /\sigma_{d_0}$	< 5	—
	$ \Delta z_0 \sin \theta $	$< 0.5 \text{ mm}$	—
	Identification	TightLH	—
	Isolation	Loose_VarRad	HighPtCaloOnly
Muon	p_T	$> 10 \text{ GeV}$	—
	$ \eta $	< 2.5	—
	$ d_0 /\sigma_{d_0}$	< 3	—
	$ \Delta z_0 \sin \theta $	$< 0.5 \text{ mm}$	—
	Identification	Loose	Medium
	Isolation	Loose_VarRad	Tight_VarRad
Hadronic tau	$ \eta $	< 1.37 or $1.52 < \eta < 2.5$	—
	Tracks	1 or 3	—
	Charge	$ Q = 1$	—
	Energy scale	MVA TES	—
	p_T	$> 20 \text{ GeV}$	—
	Jet rejection	RNN, Loose	RNN, Tight
	Electron rejection	RNN, Loose	—
Jet	Algorithm	Anti- k_t	—
	Radius	$R = 0.4$	—
	Inputs	PFlow	—
	Cleaning	LooseBad	—
	p_T	$> 20 \text{ GeV}$	—
	$ \eta $	< 2.5	—
	NNJVT	FixedEffPt (where applicable)	—
	fJVT	Not applied	—

Jets passing the GN2v01 85% working point are classified as b -jets.

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